Hypothesis 1 - Effect of Emotional Condition on Working-Memory Performance

Confirmatory Hypothesis Report

Hypothesis

Positive, negative, and neutral emotional conditions differentially affect working-memory performance.

Method

A trial-level confirmatory analysis was conducted to evaluate whether working-memory performance differed across neutral, positive, and negative emotional conditions. The analytical sample included 23 participants.

Response accuracy was analyzed using a generalized estimating equation (GEE) model with a binomial distribution, logit link, exchangeable within-participant correlation structure, and robust standard errors. Emotional condition was entered as the primary predictor, with the neutral condition specified as the reference category. The accuracy model included 17,820 trial-level observations clustered within 23 participants.

Reaction time was analyzed among valid correct-response trials only. Because reaction-time data were positively skewed, reaction time was log-transformed prior to modeling. A linear mixed-effects model was fitted with emotional condition as a fixed effect and a participant-specific random intercept. The reaction-time analysis included 14,184 observations from 23 participants. Maximum-likelihood estimation was used.

Model coefficients from the binomial GEE were exponentiated and interpreted as odds ratios. Coefficients from the log reaction-time model were exponentiated and converted to percentage changes relative to the neutral condition. Statistical significance was evaluated using a two-sided alpha level of .05, and 95% confidence intervals are reported.

Results

Accuracy

Descriptively, accuracy was highly similar across emotional conditions. Mean trial-level accuracy was 83.8% in the negative condition, 85.0% in the neutral condition, and 85.1% in the positive condition.

The omnibus Wald test indicated that emotional condition was not significantly associated with response accuracy, chi-square(2) = 2.54, p = .281.

Relative to the neutral condition, the negative condition was associated with a small, non-significant reduction in the odds of a correct response (beta = -0.084, SE = 0.064, z = -1.31, p = .189), corresponding to an odds ratio of approximately 0.92 (95% CI [0.81, 1.04]).

The positive condition did not differ from the neutral condition (beta = 0.002, SE = 0.057, z = 0.03, p = .979), with an odds ratio of approximately 1.00 (95% CI [0.90, 1.12]).

Taken together, these findings provide no evidence that emotional condition substantially altered the probability of producing a correct response.

Table 1. Descriptive accuracy by emotional condition.

Condition	Trials	Accuracy	SD
Negative	5940	83.8%	0.368

Condition	Trials	Accuracy	SD
Neutral	5940	85.0%	0.357
Positive	5940	85.1%	0.356

Table 2. GEE estimates for response accuracy.

Contrast	OR	CI low	CI high	p
Negative vs Neutral	0.92	0.81	1.04	= 0.189
Positive vs Neutral	1.00	0.90	1.12	= 0.979

Reaction Time

In contrast to accuracy, emotional condition was strongly associated with reaction time.

Relative to the neutral condition, the negative emotional condition was associated with a significantly lower log reaction time (beta = -0.265, SE = 0.015, z = -17.29, $p < .001$, 95% CI [-0.295, -0.235]). After back-transformation, this coefficient corresponds to an estimated 23.3% reduction in reaction time relative to the neutral condition (approximately 20.9% to 25.5% lower).

The positive emotional condition was also associated with a significantly lower log reaction time compared with the neutral condition (beta = -0.285, SE = 0.015, z = -18.60, $p < .001$, 95% CI [-0.316, -0.255]). This corresponds to an estimated 24.8% reduction in reaction time relative to neutral (approximately 22.5% to 27.1% lower).

The magnitude of the positive and negative coefficients was similar, suggesting that both emotionally induced conditions were associated with faster responses compared with the neutral baseline.

Table 3. Mixed-effects model estimates for log reaction time.

Contrast	Beta	RT change	CI low	CI high	p
Negative vs Neutral	-0.265	-23.3%	-25.5%	-20.9%	< .001
Positive vs Neutral	-0.285	-24.8%	-27.1%	-22.5%	< .001

Interpretation

Hypothesis 1 was partially supported.

The emotional manipulation did not produce statistically detectable differences in response accuracy. Accuracy remained approximately 84-85% across the three emotional conditions, and the global test of emotional condition was not significant.

However, emotional condition showed a pronounced association with reaction speed. Both positive and negative emotional conditions were associated with substantially faster responses than the neutral condition, with estimated reductions in reaction time of approximately 25%.

Importantly, the faster responses under emotional induction were not accompanied by a statistically significant loss of accuracy. Thus, the results suggest that emotional induction influenced the speed component of working-memory performance more strongly than the probability of a correct response.

Because both positive and negative conditions showed effects in the same direction relative to neutral, these results do not yet demonstrate a valence-specific advantage of positive emotion or a valence-specific impairment under negative emotion. Those directional predictions should be evaluated through the planned contrasts in Hypotheses 2 and 3.

Hypothesis Assessment

Conclusion: Hypothesis 1 was partially supported. Emotional condition did not significantly affect accuracy, but both positive and negative emotional conditions were associated with substantially faster reaction times relative to the neutral condition.